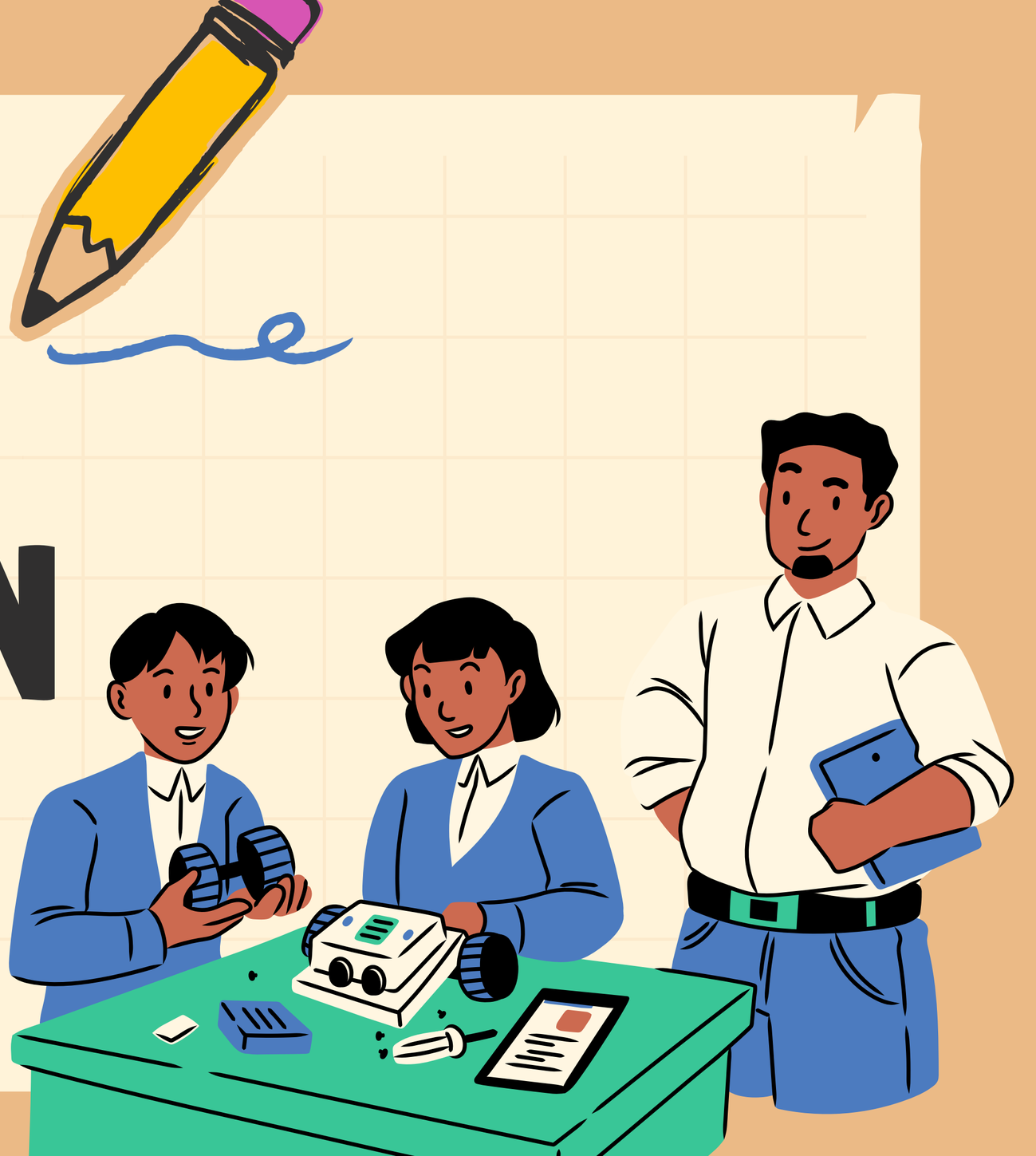
THE ATOMIC HACKATHON

Team name: Gyroscope

Team members:

- **Ayush Paul,**
- **Mayank Kumar,**
- **Priyansh Bhavsar**



INTRO

Banks face severe outreach capacity and budget caps. The challenge is to identify who to call under real-world constraints and quantify the portfolio probability of achieving target deposit conversions



ARCHITECTURE





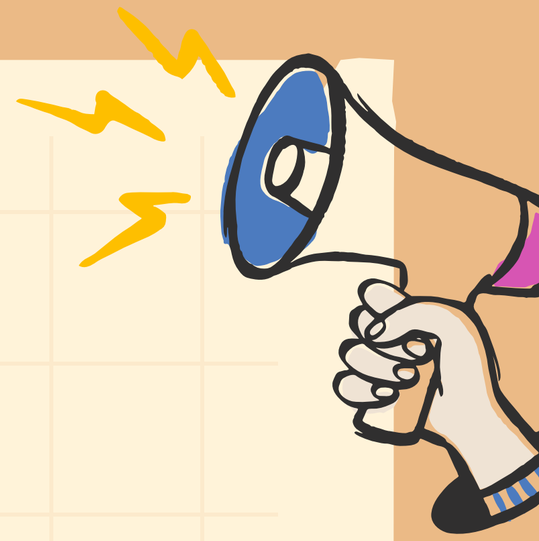
Pipeline Workflow:

- Predict: Rank 300 customers –>Top 10 candidate pool.
- Select: Constrained combinatorial optimization over 210 candidate subsets –>Optimal 6–customer cohort.
- Estimate: Multi–qubit independent Bernoulli state preparation $\rightarrow$ Portfolio risk curve for at least 3 conversions.

Architecture At A Glance:

- **Stage 1 (Predict):** Quantum-classical ensemble propensity scoring (Gradient Boosting + VQC).
- **Stage 2 (Select):** Combinatorial resource allocation via exact enumeration, QUBO, and simulated XY-mixer QAOA.
- **Stage 3 (Estimate):** Portfolio-level conversion confidence via Poisson-binomial dynamic programming and MLAE state preparation.

STAGE 1: PREDICT

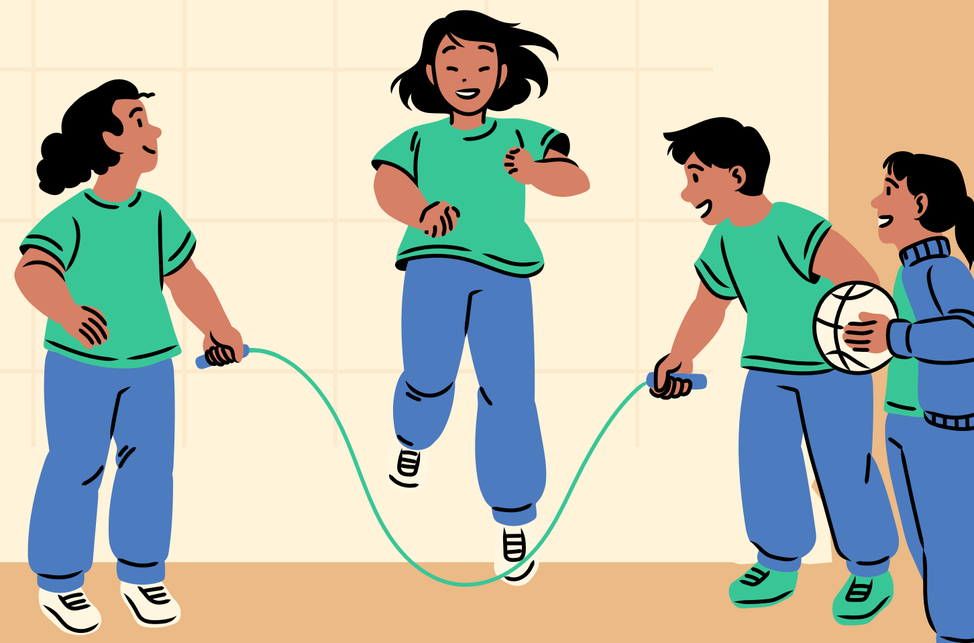


Model Exploration:

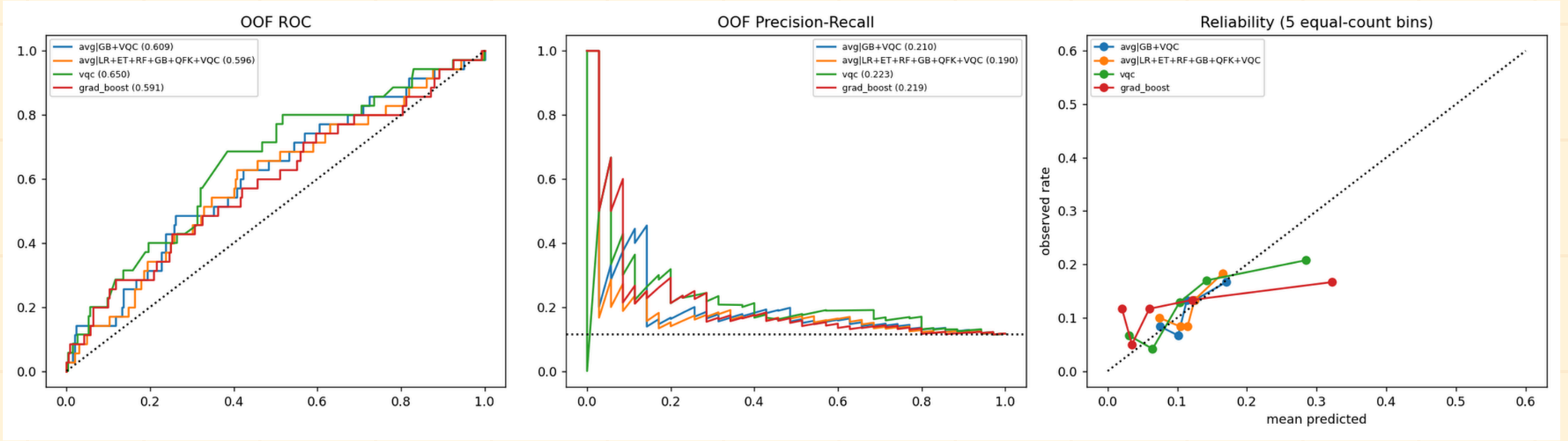
- Evaluated 4 classical models (Logistic Regression, Extra Trees, Random Forest, Gradient Boosting) against 2 simulated 6-qubit quantum models (Quantum Fidelity Kernel & Variational Quantum Classifier).
- Tested 90 mixed quantum-classical ensemble configurations.

Selected Architecture & Validation:

- Winning Model: Gradient Boosting + VQC log-odds averaging with probability calibration.
- Metrics: Cross-validation ROC-AUC of 0.618 and PR-AUC of 0.272.
- Output: Ranked top 10 candidates passed downstream, led by customer C0297 ($P = 0.5279$).



STAGE 1: RESULTS



TOP 10 customers:

customer_id	probability	rank	top10
C0297	0.527943	1	True
C0263	0.497782	2	True
C0073	0.404334	3	True
C0011	0.391731	4	True
C0107	0.382222	5	True
C0017	0.371492	6	True
C0311	0.298679	7	True
C0255	0.290453	8	True
C0151	0.290304	9	True
C0281	0.275728	10	True

STAGE 2: SELECT

Optimization Methods & Results:

- Classical Ground Truth: Exact enumeration over all 210 combinations identifies optimal bitstring 1010111001.
- Selected Cohort: C0297, C0073, C0107, C0017, C0311, C0281.
- Portfolio Metrics: Net expected value 36.9004, contact cost 18 / 25, capacity 15 / 22.

Quantum & Heuristic Benchmarks:

- XY-mixer preserves Hamming weight 6 with 1.000 valid-state probability, outperforming standard penalized X-mixers (0.573).
- Simulated annealing delivered feasible but sub-optimal value (36.7055).

$$\max \sum_{i \in S} (V_i \cdot P_i) - \text{Cost}_i \quad \text{s.t.} \quad |S| = 6,\ \text{Cost} \leq 25,\ \text{Capacity} \leq 22$$



STAGE 2: RESULTS

N=10, K=6, budget<=25.0, capacity<=22.0

	bit_index	customer_id	probability	contact_cost	expected_deposit_value	capacity_units	net_value
0	0	C0297	0.527943	4	25	4	9.198575
1	1	C0263	0.497782	2	12	1	3.973384
2	2	C0073	0.404334	2	20	1	6.086680
3	3	C0011	0.391731	3	15	1	2.875965
4	4	C0107	0.382222	4	22	2	4.408884
5	5	C0017	0.371492	3	30	1	8.144760
6	6	C0311	0.298679	3	24	3	4.168296
7	7	C0255	0.290453	3	22	1	3.389966
8	8	C0151	0.290304	4	27	1	3.838208
9	9	C0281	0.275728	2	25	4	4.893200

	bitstring	valid	value	seconds	gap_to_exact_%	matches_exact
Exact brute force (classical)	1010111001	True	36.900395	0.013872	0.000000	True
QUBO + Simulated Annealing	1110110001	True	36.705483	5.249094	0.528211	False
QAOA (XY mixer)	1010111001	True	36.900395	6.752085	0.000000	True
CVaR-QAOA (α=0.1)	1010111001	True	36.900395	4.740059	0.000000	True
Warm-Start QAOA	1010111001	True	36.900395	5.599128	0.000000	True
Grover Adaptive Search	1010111001	True	36.900395	0.002811	0.000000	True
QIEA	1010111001	True	36.900395	0.143792	0.000000	True
XY-QAOA finite-shot	1010111001	True	36.900395	0.005168	0.000000	True
XY-QAOA noisy (raw top-3)	1010111001	True	36.900395	0.001614	0.000000	True
XY-QAOA noisy + readout-mitigated	1010111001	True	36.900395	0.002847	0.000000	True
X-mixer penalised QAOA	1010111001	True	36.900395	2.419312	0.000000	True

Best solver : X-mixer penalised QAOA
Bitstring : 1010111001 (x0..x9 = top-10 rank 1..10)
Selected IDs: ['C0297', 'C0073', 'C0107', 'C0017', 'C0311', 'C0281']
Total net value = 36.900395 | exact optimum = 36.900395
Total cost = 18 (<= 25) | Total capacity = 15 (<= 22) | count = 6

12]:

	customer_id	probability	contact_cost	capacity_units	expected_deposit_value	net_value
0	C0297	0.527943	4	4	25	9.198575
2	C0073	0.404334	2	1	20	6.086680
4	C0107	0.382222	4	2	22	4.408884
5	C0017	0.371492	3	1	30	8.144760
6	C0311	0.298679	3	3	24	4.168296
9	C0281	0.275728	2	4	25	4.893200

STAGE 3: SIMULATE

For each selected customer k, the circuit applies:

$$\boxed{RY(\theta_k)|0\rangle = \cos\left(\frac{\theta_k}{2}\right)|0\rangle + \sin\left(\frac{\theta_k}{2}\right)|1\rangle}$$

The rotation angle is calculated from the customer's predicted subscription probability p_k:

$$\boxed{\theta_k = 2\arcsin(\sqrt{p_k})}$$

Therefore:

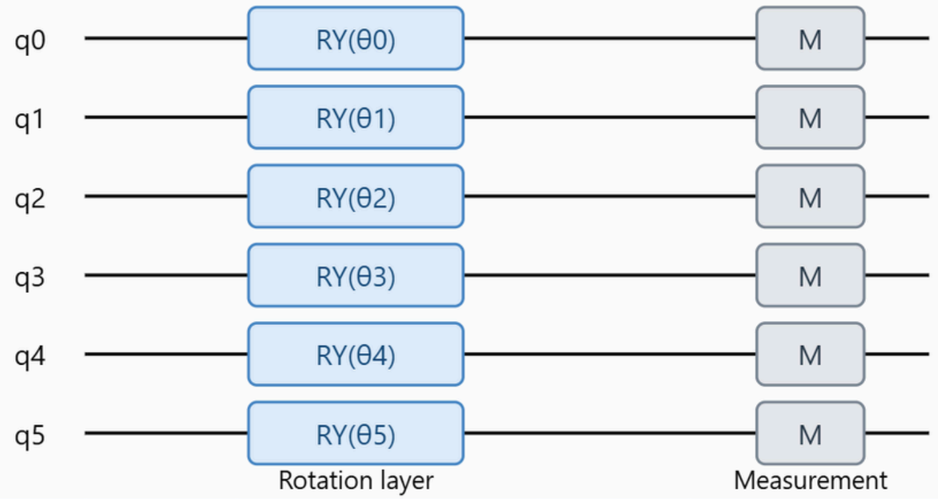
$$P(q_k = 1) = \sin^2(\theta_k/2) = p_k$$

Success condition: The objective is to estimate:

p= (X>=3)

where: (X = q_0 + q_1 + q_2 + q_3 + q_4 + q_5)

Circuit architecture



STAGE 3: RESULTS

method	regime	quantum	mean	std	bias	RMSE
Noiseless circuit shots (100k)	ideal device	True	0.725622	0.001687	-0.000155	0.001666
MLAE amplitude estimation (≈100k shots)	ideal device	True	0.725788	0.000087	0.000012	0.000086
Classical Monte-Carlo (100k trials)	classical	False	0.725662	0.001540	-0.000115	0.001518
Noisy circuit (raw)	noisy device	True	0.681950	0.001459	-0.043827	0.043850
Noisy + readout-mitigated	noisy device	True	0.724050	0.001425	-0.001726	0.002224
Noisy + readout-mitigated + ZNE	noisy device	True	0.725877	0.003112	0.000100	0.003061

Best estimator on a NOISY device : Noisy + readout-mitigated
Best estimator on an IDEAL device: MLAE amplitude estimation (≈100k shots)

	little-endian b5..b0	big-endian b0..b5	prob
0	101110	011101	0.045895
1	100110	011001	0.044095
2	111110	011111	0.037550
3	110110	011011	0.036078
4	101010	010101	0.033234
5	100010	010001	0.031931

method	P(at least 3)	error vs exact
Exact Poisson-binomial	0.725777	0.000000e+00
Ideal statevector (circuit)	0.725777	-3.330669e-16
Ideal-device best: MLAE amplitude estimation (≈100k shots)	0.725910	1.330398e-04
FINAL (noisy-device best): Noisy + readout-mitigated	0.724817	-9.593015e-04

#subscribers	exact	circuit
0	0.008387	0.008387
1	0.064946	0.064946
2	0.200890	0.200890
3	0.317660	0.317660
4	0.270508	0.270508
5	0.117389	0.117389
6	0.020219	0.020219

FINAL ANSWER: P(at least 3 of 6 subscribe) ≈ 0.724817 (exact 0.725777, |error| 9.59e-04)

QUANTUM VS. CLASSICAL DIVISION & TECHNICAL REALITY

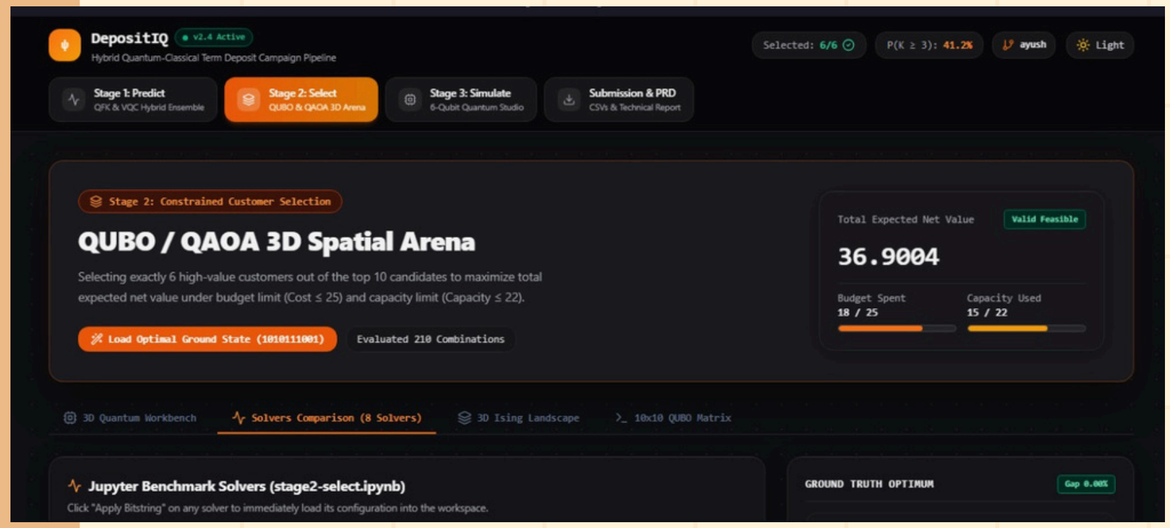
Stage	Quantum Component	Classical Component
1. Predict	6-qubit QFK & VQC variational feature maps (statevector simulation)	Data preprocessing, tree baselines, 90-ensemble search, probability calibration
2. Select	Dicke-state prep, XY-mixer QAOA, Grover adaptive search	Exact 210-subset solver, QUBO formulation, simulated annealing, COBYLA
3. Estimate	R_y state prep, circuit sampling, MLAE amplification	Poisson-binomial dynamic programming, 64-state verification, Monte Carlo

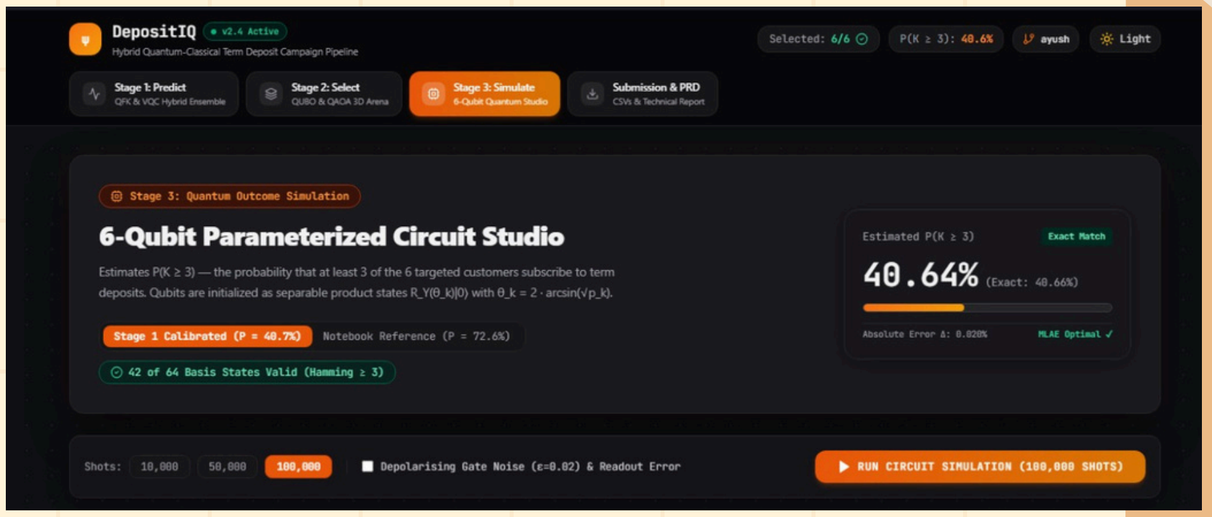
FUTURE WORKS

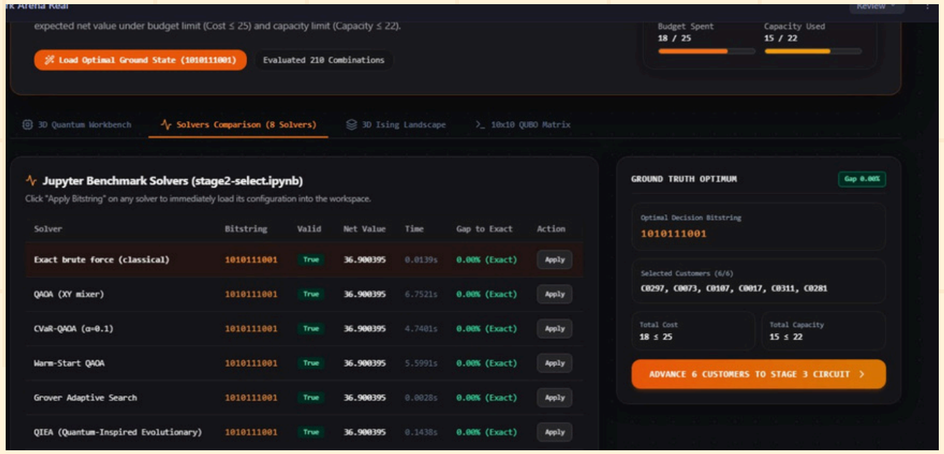
- Data Leakage & Split Integrity Fix: Re-partition data strictly along time-based out-of-time splits to prevent inflated offline metrics.
- Propensity vs. Uplift: Implement randomized controlled trials (A/B uplift modeling) to isolate true incremental lift.
- Beyond Independent Assumptions: Expand Stage 3 from unentangled product states to parameterized entangling circuits modeling correlated default/subscription events.



PROTOTYPE


DepositIQ
v2.4 Active
Hybrid Quantum-Classical Term Deposit Campaign Pipeline
Selected: 6/6
P(K ≥ 3): 41.2%
ayush
Light
Stage 1: Predict
QFK & VQC Hybrid Ensemble
Stage 2: Select
QUBO & QAOA 3D Arena
Stage 3: Simulate
6-Qubit Quantum Studio
Submission & PRD
CSVs & Technical Report
Stage 2: Constrained Customer Selection
QUBO / QAOA 3D Spatial Arena
Selecting exactly 6 high-value customers out of the top 10 candidates to maximize total expected net value under budget limit (Cost ≤ 25) and capacity limit (Capacity ≤ 22).
Load Optimal Ground State (1010111001)
Evaluated 210 Combinations
Total Expected Net Value
Valid Feasible
36.9004
Budget Spent
18 / 25
Capacity Used
15 / 22
3D Quantum Workbench
Solvers Comparison (8 Solvers)
3D Ising Landscape
10x10 QUBO Matrix
Jupyter Benchmark Solvers (stage2-select.ipynb)
Click "Apply Bitstring" on any solver to immediately load its configuration into the workspace.
GROUND TRUTH OPTIMUM
Gap 0.00%


DepositIQ
v2.4 Active
Hybrid Quantum-Classical Term Deposit Campaign Pipeline
Selected: 6/6
P(K ≥ 3): 40.6%
ayush
Light
Stage 1: Predict
QFK & VQC Hybrid Ensemble
Stage 2: Select
QUBO & QAOA 3D Arena
Stage 3: Simulate
6-Qubit Quantum Studio
Submission & PRD
CSVs & Technical Report
Stage 3: Quantum Outcome Simulation
6-Qubit Parameterized Circuit Studio
Estimates P(K ≥ 3) — the probability that at least 3 of the 6 targeted customers subscribe to term deposits. Qubits are initialized as separable product states R_Y(θ_k)|0⟩ with θ_k = 2 · arcsin(√p_k).
Stage 1 Calibrated (P = 40.7%)
Notebook Reference (P = 72.6%)
42 of 64 Basis States Valid (Hamming ≥ 3)
Estimated P(K ≥ 3)
Exact Match
40.64% (Exact: 40.66%)
Absolute Error Δ: 0.020%
MLAE Optimal ✓
Shots:
10,000
50,000
100,000
Depolarising Gate Noise (ε=0.02) & Readout Error
RUN CIRCUIT SIMULATION (100,000 SHOTS)


expected net value under budget limit (Cost ≤ 25) and capacity limit (Capacity ≤ 22).
Load Optimal Ground State (1010111001)
Evaluated 210 Combinations
Budget Spent
18 / 25
Capacity Used
15 / 22
3D Quantum Workbench
Solvers Comparison (8 Solvers)
3D Ising Landscape
10x10 QUBO Matrix
Jupyter Benchmark Solvers (stage2-select.ipynb)
Click "Apply Bitstring" on any solver to immediately load its configuration into the workspace.
Solver
Bitstring
Valid
Net Value
Time
Gap to Exact
Action
Exact brute force (classical)
1010111001
True
36.900395
0.0139s
0.00% (Exact)
Apply
QAOA (XY mixer)
1010111001
True
36.900395
6.7521s
0.00% (Exact)
Apply
CVaR-QAOA (α=0.1)
1010111001
True
36.900395
4.7401s
0.00% (Exact)
Apply
Warm-Start QAOA
1010111001
True
36.900395
5.5991s
0.00% (Exact)
Apply
Grover Adaptive Search
1010111001
True
36.900395
0.0028s
0.00% (Exact)
Apply
QIEA (Quantum-Inspired Evolutionary)
1010111001
True
36.900395
0.1438s
0.00% (Exact)
Apply
GROUND TRUTH OPTIMUM
Gap 0.00%
Optimal Decision Bitstring
1010111001
Selected Customers (6/6)
C0297, C0073, C0107, C0017, C0311, C0281
Total Cost
18 ≤ 25
Total Capacity
15 ≤ 22
ADVANCE 6 CUSTOMERS TO STAGE 3 CIRCUIT

THANK YOU!

